

FAULT CODE 235 - Coolant Level - Data Valid But Below Normal Operating Range - Most Severe Level

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Check for Fault Code 235.	Active or inactive counts of Fault Code 235?
	STEP 1B. Check the coolant level.	Coolant level normal?
STEP 2.	Check the engine harness and ECM.	
	STEP 2A. Inspect the engine harness and ECM connector pins.	Dirty or damaged pins?
	STEP 2B. Check the coolant level sensor 1 and engine harness connector pins.	Dirty or damaged pins?
	STEP 2C. Check for an open circuit in the engine harness.	Less than 10 ohms?
	STEP 2D. Check for a pin-to-pin short circuit in the engine harness.	Greater than 100k ohms?
STEP 3.	Clear the fault codes.	
	STEP 3A. Disable the fault code.	Fault Code 235 inactive?
	STEP 3B. Clear the inactive fault codes.	All fault codes cleared?

STEP 1. Check the fault codes.

STEP 1A. Check for Fault Code 235.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check for fault codes. • Use INSITE™ electronic service tool to read the fault codes.	Active or inactive counts of Fault Code 235? YES	1B
	Active or inactive counts of Fault Code 235? NO	Repair complete

STEP 1B. Check the coolant level.**⚠ WARNING ⚠**

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

Conditions :

- Engine coolant temperature below 50°C [122°F].

Action	Specification/Repair	Next Step
Check the coolant level in the radiator top tank or surge tank. Make sure the engine coolant temperature is below 50°C [122°F] before removing the radiator cap.	Coolant level normal? YES	2A
	Coolant level normal? NO Repair : Fill the radiator top tank or surge tank with coolant.	3A

STEP 2. Check the engine harness and ECM.**STEP 2A. Inspect the engine harness and ECM connector pins.**

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM connector.

Action	Specification/Repair	Next Step
Inspect the engine harness and ECM connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the engine harness or ECM connector. Clean the connector and pins. Replace the damaged section of the engine harness or the ECM. Check all harnesses connected in series. Refer to the circuit diagram or wiring diagram for all harness interconnections. • Repair the harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html) • Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)	3A
	Dirty or damaged pins? NO	2B

STEP 2B. Inspect the coolant level sensor 1 engine harness and the connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the coolant level sensor 1 connector.

Action	Specification/Repair	Next Step
Inspect the engine harness and the coolant level sensor 1 connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the engine harness or the coolant level sensor 1 connector. Clean the connector and pins. Replace the damaged section of the engine harness or the the coolant level sensor 1. Check all harnesses connected in series. Refer to the circuit diagram or wiring diagram for all harness interconnections. • Repair the harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html) • Replace the coolant level sensor 1. Refer to Procedure 019-017 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-017.html)	3A
	Dirty or damaged pins? NO	2C

STEP 2C. Check for an open circuit in the engine harness.

Conditions :

- Turn keyswitch OFF.
- Disconnect the coolant level 1 sensor connector from the engine harness.
- Disconnect the engine harness from the ECM connector.

Action	Specification/Repair	Next Step
Check for an open circuit. • Measure the resistance between the coolant level 1-5 volt supply (sensor supply1) SUPPLY pin of the coolant level sensor 1 engine harness connector and the coolant level 1-5 volt supply (sensor supply 1) SUPPLY pin of the engine harness ECM connector. Refer to the circuit diagram or the wiring diagram for connector pin identification. Refer to the circuit diagram or wiring diagram for connector pin identification. Refer to Procedure 019-360 in Section 19 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Less than 10 ohms? YES	2D
	Less than 10 ohms? NO Repair : An open SUPPLY wire has been detected in the engine harness. Troubleshoot all harnesses connected in series to determine which contains the open circuit. Refer to the circuit diagram or wiring diagram for all harness interconnections. Repair the harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)	3A

STEP 2D. Check for a pin-to-pin short circuit in the engine harness.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM connector.

Action	Specification/Repair	Next Step
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Check for a pin-to-pin short circuit. Measure the resistance between the coolant level 1 SIGNAL pin and the battery 1 voltage (ECM1) pin of the ECM 4-pin OEM power harness. Refer to the circuit diagram or the wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES Repair : Return to the troubleshooting steps. A fault should have been detected.	1A
	Greater than 100k ohms? NO Repair : A pin-to-pin short circuit has been detected in the coolant level 1 SIGNAL wire of the engine harness. Troubleshoot all harnesses connected in series to determine which contains the pin-to-pin short circuit. Replace the damaged section of the harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html)	3A

STEP 3. Clear the fault codes.

STEP 3A. Disable the fault code.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Disable the fault code. Use INSITE™ electronic service tool to verify the fault code is inactive.	Fault Code 235 inactive? YES	2B
	Fault Code 235 inactive? NO Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A

STEP 2B. Clear the inactive fault codes.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Clear the inactive fault codes. Use INSITE™ electronic service tool to clear the inactive fault codes.	All fault codes cleared? YES	Repair complete
	All fault codes cleared? NO Repair : Troubleshoot any remaining fault codes.	Appropriate troubleshooting steps